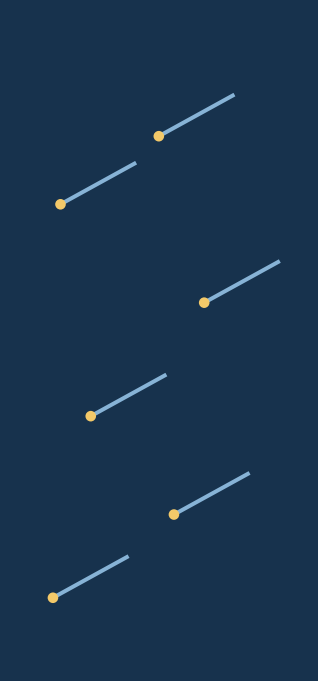




How should the planner predict future camera observations?

Separate method explanations

Supervisor discussion — Joost Leliveld — 10 August 2026

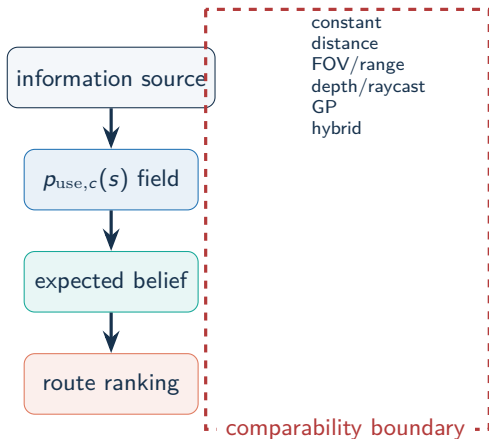
Presentation contract: every visual in this deck is a new schematic. No old run, plot, fit or numerical result is used as evidence.

One target, several information sources

$$p_{\text{use},c}(s)$$

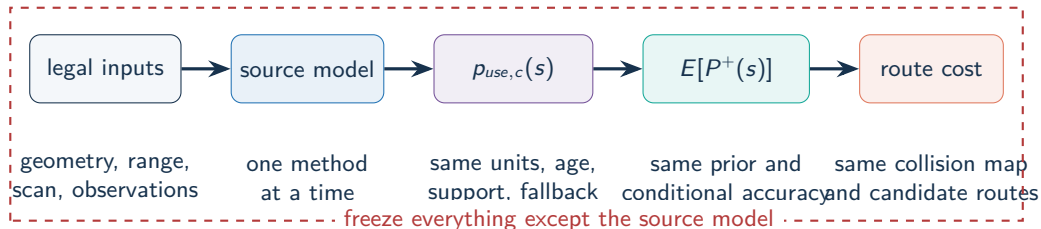
is the probability that camera c yields a usable observation at candidate pose s .

- ▶ Every source predicts the **same quantity**.
- ▶ Planner, robot, routes and belief update stay fixed.
- ▶ Only information available at a **future candidate pose** is legal.
- ▶ Current detector outcome and simulator truth are evaluation-only.



Takeaway: A fair comparison changes the source of p_{use} , not the planner around it.

Shared pipeline: source $\rightarrow$ field $\rightarrow$ decision



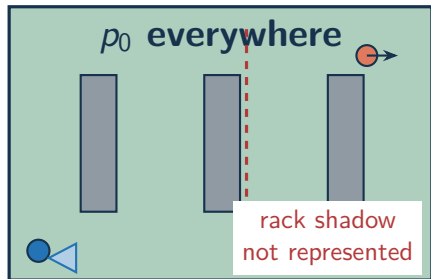
Two updates must stay separate

Runtime observations update the **robot belief**. Only learned/maintained sources update the **reliability field**. A detector hit is not automatically a map update.

Method 1 — Conservative constant

- ▶ **Knows:** one prevalence p_0
- ▶ **Outputs:** the same p_{use} everywhere
- ▶ **Updates:** only at explicit recommissioning
- ▶ **Fails when:** spatial structure matters

$$p_{use,c}(s) = p_0$$

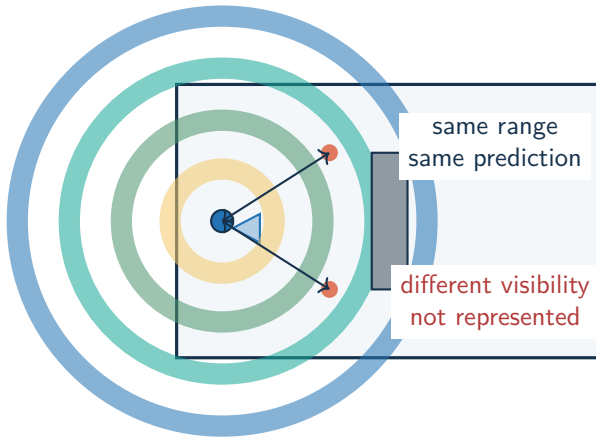


Takeaway: This is a necessary null model, not a straw man: complexity must beat it on held-out blocks.

Method 2 — Distance-only

- **Knows:** camera position and a distance curve
- **Outputs:** smooth radial reliability
- **Updates:** when the curve is recommissioned
- **Fails when:** equal-range poses differ through orientation or occlusion

$$p_{use,c}(s) = f(\|s - c\|), \quad f' \leq 0$$

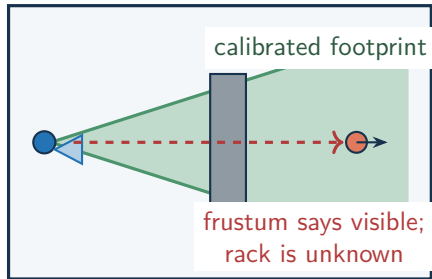


Takeaway: Distance earns its place by revealing whether scene-aware inputs are actually necessary.

Method 3 — Calibrated FOV and range

- **Knows:** camera pose, intrinsics, image bounds, target height
- **Outputs:** analytic in-frustum/range score
- **Updates:** after calibration changes
- **Fails when:** an obstacle blocks an otherwise valid ray

$$p_{use,c}(s) = g(\text{in-FOV}, \text{range}, \text{obliquity})$$

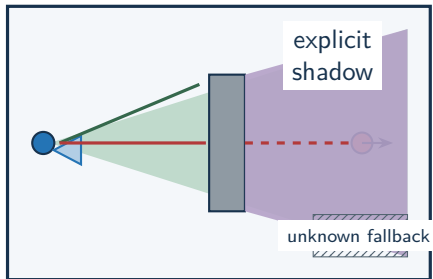


Takeaway: FOV/range adds camera geometry, but not scene occlusion.

Method 4 — Sensed depth and raycast

- ▶ **Knows:** camera calibration, sensed 3-D surface, target height, scan age
- ▶ **Outputs:** line-of-sight reliability with unknown-cell fallback
- ▶ **Updates:** on rescan or declared live-depth update
- ▶ **Fails when:** depth is missing, stale or misregistered

depth → point cloud → height map → raycast

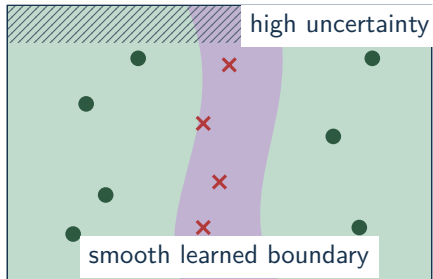


Takeaway: Depth provides cold-start occlusion structure, but it creates a freshness and missing-data obligation.

Method 5 — Operational Gaussian process

- **Knows:** pose-labelled usable/miss blocks for each camera
- **Outputs:** posterior mean plus separate epistemic support
- **Updates:** when eligible operational blocks arrive
- **Fails when:** sharp boundaries or unsupported space are smoothed/extrapolated

$$f_c(s) \sim \mathcal{GP}(m_c, k_c), \quad p_{use} = \sigma(f_c)$$

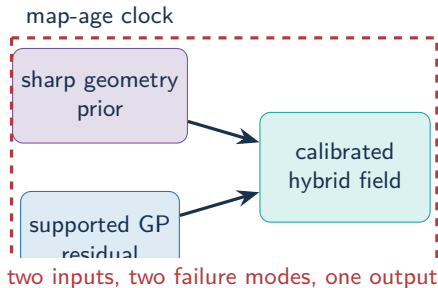


Takeaway: The GP can learn installed-view effects, but repeated frames do not create new spatial support.

Method 6 — Hybrid geometry plus learned residual

- **Knows:** depth/raycast prior plus operational blocks
- **Outputs:** calibrated geometric prior corrected by a supported residual
- **Updates:** residual on observations; geometry on rescan
- **Fails when:** a stale confident prior overwhelms sparse contradictory data

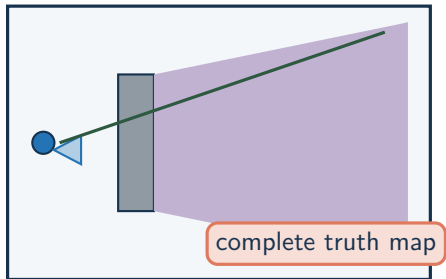
$$\text{logit } p_{use} = \text{logit } p_{depth} + r_{GP}$$



Takeaway: Freeze the candidate equation and reset rule first; hybrid must then beat both parents at equal budget.

Reference — Complete CAD raycast

- ▶ **Knows:** complete simulator/SDF geometry and exact registration
- ▶ **Outputs:** full static line-of-sight reference
- ▶ **Updates:** when oracle world truth changes
- ▶ **Fails when:** its inputs are unavailable or stale in deployment



EVALUATION-ONLY ORACLE

Cannot win an operational comparison.

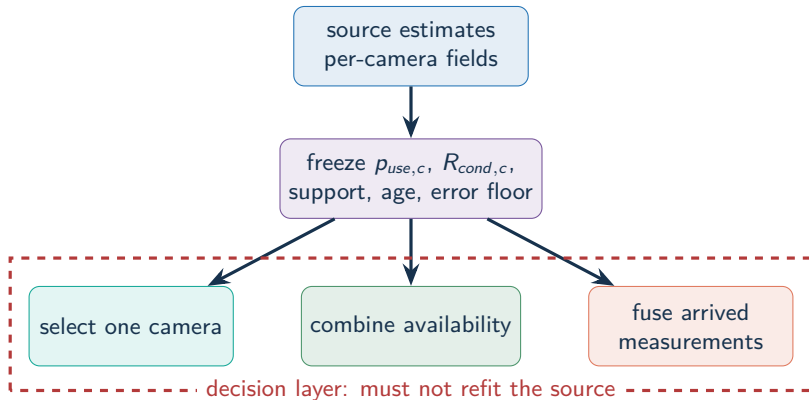
Takeaway: CAD measures the geometric ceiling and diagnoses operational methods; it is not free deployment knowledge.

Source methods side by side

Method	Starts with	Changes when	Characteristic failure
Constant	prevalence	recommissioned	no spatial structure
Distance	camera position + curve	curve is refitted	ignores orientation and occlusion
FOV/range	calibrated projection	camera is recalibrated	sees through obstacles
Depth/raycast	sensed 3-D scan	geometry is rescanned	missing, stale or misregistered geometry
Operational GP	prior + usable/miss blocks	new independent blocks arrive	smooth extrapolation in unsupported space
Hybrid	depth prior + residual model	either input clock changes	stale prior conflicts with learned residual
CAD reference	complete simulator truth	oracle world changes	non-operational input

Takeaway: The comparison is about information, maintenance and failure modes—not only predictive score.

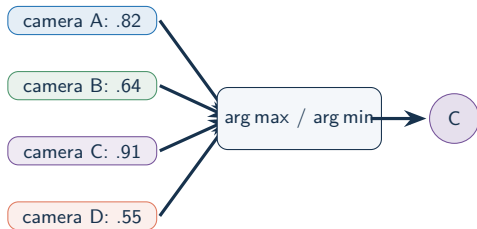
Do not mix estimation with camera management



Takeaway: A source can be good while a downstream selection/fusion policy is poor, and vice versa.

Decision method A — Select one camera

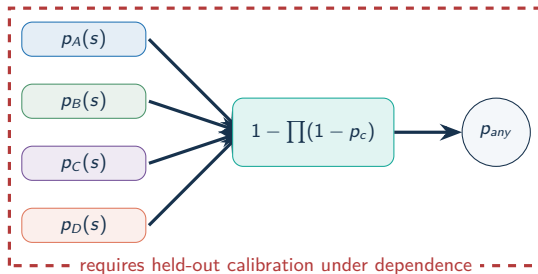
- **Knows:** frozen per-camera fields
- **Outputs:** one camera identity per candidate pose
- **Updates:** when health or candidate pose changes
- **Fails when:** the chosen scalar ignores accuracy, correlation or switching cost
- max availability: $\arg \max_C p_{use,c}$
- achievable precision:
 $\arg \min_C \sigma_{achievable,c}$
- hysteresis prevents rapid switching



Takeaway: Selection is interpretable and correlation-safe, but discards simultaneous complementary measurements.

Decision method B — Combine availability fields

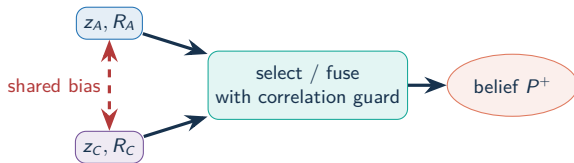
- **Knows:** frozen per-camera usable-observation probabilities
- **Outputs:** probability that at least one usable observation arrives
- **Updates:** when a component field/health changes
- **Fails when:** camera events are treated as independent when they are correlated



Takeaway: Noisy-OR is a probability rule, not measurement fusion; dependence must earn held-out calibration.

Decision method C — Fuse arrived measurements

- **Knows:** actual time-aligned camera measurements and covariances
- **Outputs:** one state correction
- **Updates:** at each realized measurement event
- **Fails when:** shared systematic error is counted repeatedly
- best-camera selection;
- naive independent sequential fusion;
- health-aware suppression / conservative floor.

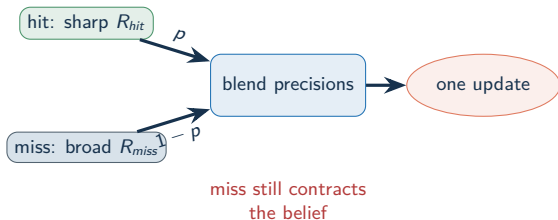


Takeaway: More measurements do not guarantee more information when their errors share a cause.

Planner approximation 1 — Precision blend

- **Knows:** p_{use} plus visible/miss covariance endpoints
- **Outputs:** one effective measurement covariance
- **Updates:** at every candidate state
- **Fails when:** a miss is represented as a weak measurement instead of no measurement

$$R_{blend}^{-1} = pR_{hit}^{-1} + (1 - p)R_{miss}^{-1}$$



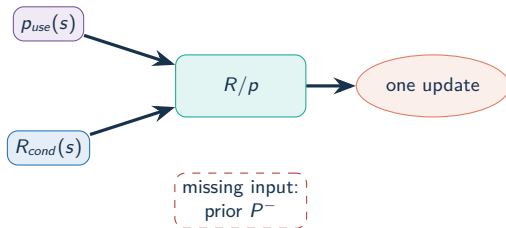
Takeaway: Computationally simple, but it collapses a discrete hit/miss event into one Gaussian promise.

Planner approximation 2 — Scale covariance by $1/p$

- **Knows:** p_{use} and conditional covariance R
- **Outputs:** one inflated covariance R/p
- **Updates:** at every candidate state
- **Fails when:** the equivalent covariance also depends on the prior belief

$$R_{eff} = R_{cond} / p_{use}$$

No pose-only cached $R_{eff}(s)$ can be exact for all priors.



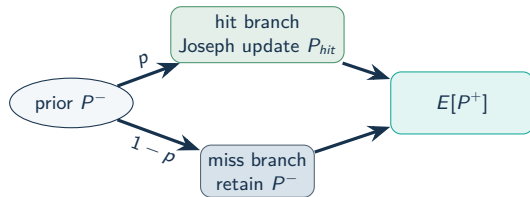
Takeaway: R/p is endpoint-correct but structurally cannot reproduce the expected branch for every prior.

Planner method 3 — Explicit expected hit/miss branch

- **Knows:** p_{use} , R_{cond} and prior P^-
- **Outputs:** expected posterior covariance
- **Updates:** for every candidate state and current prior
- **Fails when:** the source probabilities or conditional covariance are miscalibrated

Rule: $E[P^+] = p_{use}P_{hit} + (1 - p_{use})P^-$.

A miss performs **no update**.



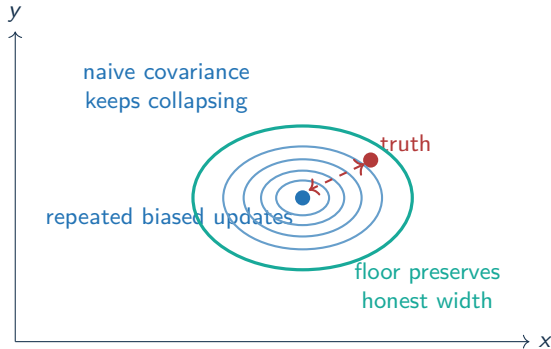
Takeaway: This preserves the event semantics: usable observation or no observation.

Belief containment — Persistent-error correlation floor

- **Knows:** fresh held-out residual blocks grouped by camera/site
- **Outputs:** a lower bound on measurement uncertainty plus cross-camera check
- **Updates:** only after fresh commissioning and lifecycle validation
- **Fails when:** the floor is learned from confounded or stale residuals

$$R_{used,c}(s) \succeq R_{floor,c}$$

This is containment, not proof that the bias source is camera calibration.



Takeaway: The floor prevents false confidence; fresh identifiability is still required before attributing its source.

What is fresh, and what is not evidence yet?

This deck

- ▶ New vector schematics only
- ▶ No old captures, runs, fits or plots
- ▶ Explains mechanisms and decision boundaries
- ▶ Makes no ranking or navigation claim

Fresh evidence still required

- ▶ 6 sites $\times$ 8 yaws $\times$ 2 sessions
- ▶ 96 capture blocks; A–D simultaneous
- ▶ site-grouped analysis, never frame inference
- ▶ fresh source benchmark on shared blocks
- ▶ complete manifests and hashes

method explanation

fresh collection

freeze analysis

bounded claim/null

Decisions requested from the supervisor

1. Is $p_{use,c}(s)$ the right common estimand and label definition?
2. Which depth provenance is the primary operational arm: commissioning scan, live sensed depth, or commissioned static map?
3. Is the hybrid candidate form acceptable, and what reset/decay rule follows a rescan?
4. Approve or amend the fresh 6-site $\times$ 8-yaw $\times$ 2-session identifiability design and its practical-effect/harm gates.
5. Which single endpoint should govern a later matched closed-loop campaign, if the identifiability gate passes?

Proposed meeting outcome: freeze the comparison contract, not a winner.